James Liu

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Education

Stanford University — Stanford, CA
B.S., M.S. Computer Science, Mathematics — *May 2028*

- Relevant Coursework: Compilers, Networking, OS, Machine Learning, NLP w/ DL, Deep RL, CV w/ DL, Linear Algebra and Matrix Theory, Probability Theory, Measure Theory & Stochastic Processes, Cloud Computing, Algorithms
- Activities: TreeHacks Organizer (Hackathon w/ 1,000+ Participants), AI Club, Symphony Orchestra, Math Club

Experience

OpenAI — Jun 2026-
Incoming Member of Technical Staff Intern — *San Francisco, CA*

- Incoming Summer 2026.

Stanford Artificial Intelligence Lab | Professor Yejin Choi — Jun 2025-Present
Research Assistant — VERL, vLLM, Slurm — *Stanford, CA*

- Built a **multi-agent** RL pipeline using VERL with Slurm-managed training on NVIDIA H100s; trained shared-parameter LMs using dual reward functions for quality and diversity, improving Shannon Evenness Index by over **15%**.

Stanford School of Engineering (CS198) — Apr 2025-Present
Teaching Assistant — *Stanford, CA*

- Lead sections of 15 students for Stanford's DSA class. Grade assignments and debug Python & C++ code.

MultiOn - AGI, Inc — Jan 2025-Apr 2025
AI Engineer Intern — LangChain, LangGraph, Chrome DevTools — *San Francisco & Stanford, CA*

- Designed a benchmark for evaluating WebAgents and prototyped a memory-augmented browser agent using LangChain, LangGraph, and OpenAI APIs; fine-tuned the agent with RLHF via GRPO, **over 10%** reduction in average latency.

Stanford Carta — Dec 2024-May 2025
Backend Engineer — Docker, C#, SQL, .NET Core — *Stanford, CA*

- Reduced p95 API latency 30% across 4 .NET Core services that power the web app (∼2k MAU) by adding read-through Redis caching and tuning SQL connection pools
- Built a Dockerized SQL ingestion pipeline handling around 100k records/week (bulk upserts, idempotency, retry/backoff) with 99.9% success and sub-second median latency

Jane Street Capital — Jun 2024-Aug 2024
Academy of Math and Programming — *Manhattan, NY*

- Applied **optimization theory** and time series data to develop trading software using Python and statistical models
- Analyzed problems in discrete mathematics, probability, and theoretical computer science, designing stochastic models.
- Won 1st place in the Mystery Planet Puzzle Hunt and 2nd in Trading Competition; awarded $5,000 merit scholarship.

Projects / Research

Agent Benchmarking and Architecture (NeurIPS Poster 2025) — Jan 2025-Mar 2025

- Co-authored *REAL*, a benchmark for evaluating autonomous web agents across browsing, form-filling, and task completion scenarios; defined evaluation metrics for reliability, accuracy, and latency.

Market-Making Bot | *Python, AWS Lambda, WebSockets* — June 2025-Present

- Developing an automated market-making bot for prediction markets like Kalshi event contracts, streaming real-time order book data via WebSockets and executing trades through REST APIs with inventory-risk controls.

SV Hub Hackathon Winner ($50k top prize) – Pitched to Eric Schmidt, Sebastian Thrun, etc — Jan 2025

- Developed a multi-agent RAG pipeline using fine-tuned LLaMA-2 and Reflexion agents to automate O-1 visa validation by parsing legal documents and extracting expert evidence criteria with high accuracy.

Jane Street Electronic Trading Challenge Algorithm (2nd/80+) | *Python, AWS EC2* — Jul 2024

- Developed low-latency trading strategies in Python for Jane Street's Electronic Trading Challenge, including ADR/ETF arbitrage, mean reversion, and market making; generated over $10M PnL in simulation against 80+ teams.

Additional Information

Languages: Java, Python, C/C++, C#, Assembly (x86-64), SQL (Postgres), JavaScript, HTML/CSS
Frameworks: React, Node.js, Flask, FastAPI, JUnit
Developer Tools & Libraries: Pytorch, Tensorflow, Git, Docker, Kubernetes, MongoDB, AWS, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse, pandas, NumPy, Matplotlib, GDB, Valgrind, Vim, HuggingFace, Scikit-learn
Competitions and Awards: QRT Poker Tournament 3rd/100, Citadel Trading Invitational, GreylockX Fellow, TreeHacks 2026 Best Multi-Turn Agent Winner, SIG Inter-Collegiate Poker Tournament 3rd/1000+; $4k, Paradigm Fellow
Interests: Tennis, Marathon, Poker, Board Games (Figgie, Coup), Juilliard Pre-College, NYO-USA, NYYS (Grammy-Winner)